Scalable Probit Calibration

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Abstract

Probit models are a cornerstone of economic, choice, and contest theory, and machine learning runs on the same latent-utility family. Yet their many-alternative form is barely used, because probit probabilities have no closed form and the standard simulator prices one alternative at a time. Calibration, recovering the utilities at which model shares match observed shares, has stayed out of reach beyond small menus. We give an algorithm that calibrates factor-structured probit at a thousand alternatives in under a minute. Conditionally on the factors the alternatives are independent, so one shared survival product yields every choice probability in a single pass, differentiable without resampling. The inverse problem is well posed: shares determine utilities up to a common shift. Forward errors stay below one part in a thousand from five to five thousand alternatives. The method requires a factor approximation of the covariance, and the experiments locate where that fails.

1 Introduction

Probit models are a cornerstone of economic, choice, and contest theory: Thurstone's comparative judgment, the ordered and binary probits of applied microeconometrics, and the Lazear–Rosen tournament are all the same Gaussian latent-performance model. Machine learning runs on the same family: a softmax output layer is its Luce–Gumbel member, and Thurstone–Mosteller pairwise comparison, the probit member, underlies rating systems and preference learning.

Such models are calibrated to observed shares so that the recovered utilities can answer questions the data do not: elasticities, welfare, pricing, the effect of adding or removing an alternative. The models used for this, essentially universally, are the logit family: multinomial logit, nested logit, and above all random-coefficients logit in the BLP tradition [11, 4, 12]. They are universal for one reason. They can be computed.

Choice modeling did not start there: it started with Thurstone's 1927 model [1] of jointly Gaussian utilities with unrestricted correlation, known in econometrics as multinomial probit (MNP).

*Numbers in this paper are produced by committed, seeded scripts with correctness anchors run before any comparison (`https://github.com/microprediction/kinetics`, experiments 13–18; the exact commit is recorded in the repository README and an immutable tag is pinned at circulation). The algorithm ships in the `thurstone` package; package–benchmark agreement is itself a reported number. Benchmarks: Apple M4, 16 GB, Python 3.12.9, NumPy 2.4.6, SciPy 1.18.0; experiment 17 enforces single-threaded BLAS via `threadpoolctl` and reports median-of-3 timings for sub-second calls; long simulations are single realizations, disclosed as such. Monte Carlo chunk sizes derive from an explicit memory budget (1.5 GB); peak resident memory stays below 2 GB.

Probit probabilities are $(N-1)$-dimensional Gaussian integrals with no closed form. The standard evaluator is the Geweke–Hajivassiliou–Keane (GHK) simulator [2, 3, 4]: it writes the probability that alternative i beats the rest as an orthant probability of the difference utilities, and estimates it by sampling truncated normals sequentially along a Cholesky factor. That is a separate simulation per alternative, with $R^{-1/2}$ error in the number of draws. So probit survives mainly on small menus, and the field kept the computable models and gave up the model it started from.

The concession is not free. Logit-family substitution is structurally restricted, and the calibration apparatus that demand estimation is built on exists on the logit side only. One measure of the cost: in the substitution study of Section 4, plain logit misallocates about 23% of the demand released by a deleted alternative, against 5% for the correctly structured probit model.

This paper's point is that for the covariance structure applied work most often actually has, the factor model $\Sigma = VV^\top + \operatorname{diag}(D)$, the barrier is not real. The goal is **calibration**: given observed shares, recover the utilities at which the probit model reproduces them, the probit analogue of the inversions demand estimation runs on the logit side every day [11, 12]. We solve it at $N = 1000$ in under a minute, and the inverse problem is globally well-posed (Theorem 1: shares determine utilities uniquely up to location). Three ingredients make it work:

1. **A fast forward map** (utilities $\rightarrow$ shares): all N probabilities at once in $O(QNL)$ for Q factor nodes and L lattice points. All 5000 shares of an $N = 5000$ problem take 21 seconds, where matched-accuracy per-alternative simulation extrapolates to roughly fifty hours. Calibration needs many forward evaluations; this is what makes them affordable.

2. **Resampling-free derivatives**: Jacobian–vector products in $O(QNL)$; the Jacobian is an exact weighted graph Laplacian, so the reduced systems Newton solves are symmetric positive definite.

3. **A single shared construction**: conditionally on the k factors the alternatives are independent, and a lattice survival-product identity computes every alternative's integral from one *shared survival field*, built once and reused for shares, slopes, and all N single-removal counterfactuals.

Prior art. The nearest methods, and what each lacks:

approach	all N shares	calibration
GHK simulation [2, 3]	N separate runs, $R^{-1/2}$ error	not attempted at scale
analytic approximations [7, 8]	per-alternative	no
factor quadrature [5, 6]	still per-alternative	no
Bayesian factor MNP [17]	simulation inside estimation	posterior, not a share map
neural amortization [14]	fast, uncontrolled error	no
BLP-side inversion [11, 12]	—	logit family only
this paper	one $O(QNL)$ pass	Thm. 1; $<$1 min at N=1000

Figure 1 shows the running-time consequence as the number of alternatives grows. The per-alternative simulator's cost crosses the lattice transform near $N \approx 200$, and at that point its error is three orders of magnitude larger (Table 1). By $N = 5000$ the lattice computes all shares in 21 seconds; matched-accuracy GHK extrapolates to roughly fifty hours.

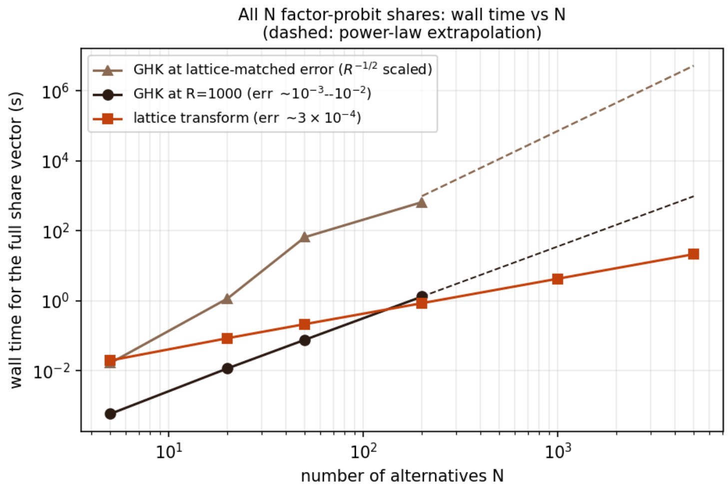

Figure 1: Wall time for the full N-alternative share vector as N grows ($k = 2$ factors). The upper GHK line rescales R by the squared error ratio ($R^{-1/2}$ scaling) so both methods deliver the same accuracy; at $N = 5000$ it extrapolates to roughly fifty hours against the lattice's 21 seconds (Section 2). Dashed segments are power-law extrapolations; all other points are measured (Table 1).

Scope. The method needs the factor form. Where Σ has no adequate low-rank-plus-diagonal approximation in the choice-relevant quotient space, simulation at exact Σ remains the tool (Section 4). At accuracy targets near 10^{-3}, plain direct utility simulation is a genuinely competitive forward baseline (Section 2). The decisive advantages are error decay below that level, reproducibility, resampling-free derivatives, and the calibration machinery.

The one-pass identity comes from an unlikely literature: inferring ability from win probabilities in multi-entrant races [10]. That is perhaps why it has not been assembled for probit before, despite each ingredient being classical.

Model. Throughout,

$$U_i \;=\; \mu_i + v_i^\top f + \sqrt{D_i}\,\varepsilon_i, \qquad f \sim N(0, I_k), \quad \varepsilon_i \sim N(0,1) \text{ i.i.d.}, \quad D_i > 0, \tag{1}$$

i.e. $\Sigma = VV^\top + \mathrm{diag}(D)$; the chosen alternative is the argmax of U. Factor dimension reduction for probit is classical: one-factor quadrature goes back to Butler and Moffitt [5], and factor-analytic probit to Elrod and Keane [6]. The contribution is the all-alternatives assembly, the inverse transform, the derivative structure, and evidence for all of it.

Proposition 1 (Conditional identity)**.** *Under* (1)*, with $g_i(\cdot \mid f)$ and $F_i(\cdot \mid f)$ the conditional density and CDF of U_i given f,*

$$p_i \;=\; \mathbb{E}_f\!\left[\int g_i(x \mid f) \prod_{j \neq i} F_j(x \mid f)\, dx\right], \qquad \sum_i p_i = 1, \tag{2}$$

and the product over $j \neq i$, for all i simultaneously, is obtained from $\log F_{\text{field}} = \sum_j \log F_j$ by subtraction.

Proof. Conditional on f the U_j are independent, so $\mathbb{P}(U_i \text{ maximal} \mid f) = \int g_i \prod_{j \neq i} F_j \, dx$; average over f. The events partition (ties are null for continuous laws), giving $\sum_i p_i = 1$. □

Proposition 2 (Complexity)**.** *With Q factor nodes and L lattice points, the forward share vector costs $O(QNL)$; the full single-removal ensemble $\{P(j \text{ chosen} \mid i \text{ removed})\}_{i,j}$ costs $O(QN^2L)$ arithmetic and $O(N^2)$ storage, reusing one conditional field construction.*

Proof. Per node: the field log-product costs $O(NL)$, each alternative's integral $O(L)$ after subtraction, giving $O(NL)$; the removal ensemble evaluates N survivor integrals for each of N removals, $O(N^2L)$, over the same cached field. □

Proposition 3 (Identification and differentiability)**.** *$p(\mu + c\mathbf{1}) = p(\mu)$, so $J = \partial p/\partial\mu$ has $\mathbf{1}$ in its null space and μ is identified only up to location. With B an $N \times (N-1)$ basis of $\mathbf{1}^\perp$ and $\mu = B\eta$, share inversion and implicit differentiation are taken in reduced coordinates: $d\eta/d\theta = -(B^\top JB)^{-1}B^\top \partial p/\partial\theta$, provided $B^\top JB$ is nonsingular.*

Proof. Invariance holds because a common location shift does not change the argmax; differentiating $p(\mu + c\mathbf{1}) = p(\mu)$ in c gives $J\mathbf{1} = 0$. Differentiating $p(B\eta(\theta), \theta) = \hat{p}$ in θ and projecting with $B^\top$ gives the reduced formula under the stated regularity. □

Theorem 1 (Shares determine utilities)**.** *For model* (1) *with $D_i > 0$, in max-wins coordinates, define for $i \neq j$*

$$w_{ij} \;=\; \mathbb{E}_f \int g_i(x \mid f)\, g_j(x \mid f) \prod_{\ell \neq i,j} F_\ell(x \mid f)\, dx \;>\; 0.$$

Then $J = \partial p/\partial\mu$ satisfies $J_{ij} = -w_{ij}$ and $J_{ii} = \sum_{j \neq i} w_{ij}$, i.e.

$$J \;=\; \sum_{i<j} w_{ij}(e_i - e_j)(e_i - e_j)^\top, \qquad h^\top Jh \;=\; \sum_{i<j} w_{ij}(h_i - h_j)^2,$$

a weighted graph Laplacian with everywhere-positive weights: J is symmetric, positive semidefinite, has $\mathbf{1}$ as its only null direction, and $B^\top JB \succ 0$ for every full-rank basis B of $\mathbf{1}^\perp$. In English: w_{ij} is the probability density of a photo-finish between i and j, and moving utility from j to i moves share along every photo-finish edge at exactly that rate. Moreover $p = \nabla G$ for the social-surplus function $G(\mu) = \mathbb{E}\max_i\{\mu_i + \xi_i\}$, and the restriction of p to $\mathbf{1}^\perp$ is a smooth bijection onto the interior of the probability simplex: every strictly positive target $\hat{p}$ has exactly one mean-zero utility vector.

Proof. Differentiating (2) in μ_j and integrating by parts gives $J_{ij} = -w_{ij}$ for $i \neq j$; the diagonal follows from $J\mathbf{1} = 0$ (Proposition 3), and positivity of w_{ij} from positivity of Gaussian densities. The quadratic form is then immediate, and it vanishes only for constant h. For inversion, minimize $H(\mu) = G(\mu) - \hat{p}^\top\mu$ over $\mathbf{1}^\perp$: H is strictly convex there (its Hessian is the reduced J), and since the shocks have zero mean, $G(\mu) \geq \max_i \mu_i$, so

$$H(\mu) \;\geq\; \max_i \mu_i - \hat{p}^\top\mu \;=\; \textstyle\sum_i \hat{p}_i(\max_j \mu_j - \mu_i) \;\geq\; \hat{p}_{\min}\,(\max_i \mu_i - \min_i \mu_i) \;\to\; \infty$$

as $\|\mu\| \to \infty$ on $\mathbf{1}^\perp$. The unique minimizer satisfies $\nabla G = \hat{p}$. □

Remark 1. Social-surplus convex duality for discrete choice is classical [15, 16]; what is new here is not well-posedness but its scalable realization. Each identity in the theorem is also checked numerically in the repository: the w_{ij} representation against finite differences to 6×10^{-9}, symmetry to 4×10^{-12}, and positive definiteness of the reduced numerical Jacobian.

Proposition 4 (Jacobian–vector products in $O(QNL)$)**.** *With hazards $\lambda_j = g_j/F_j$, $\Lambda = \sum_j \lambda_j$, and $A = \sum_j h_j\lambda_j$ (all conditional on f, all functions of x), the Jacobian of the exact max-wins map acts on a direction h as*

$$(Jh)_i \;=\; \mathbb{E}_f \int g_i\Big(\prod_{j\neq i} F_j\Big)\,(h_i\Lambda - A)\,dx,$$

where the field sums Λ and A are formed once per node and lattice point, so all N components cost $O(NL)$ per node. In English: a direction h perturbs alternative i's share only through the gap between its own hazard weight $h_i\Lambda$ and the field average A. Translation invariance is visible ($h = \mathbf{1}$ gives $A = \Lambda$, hence $Jh = 0$), and the weighted-Laplacian form of Theorem 1 follows by expanding A. This enables Krylov solves of the reduced system in Proposition 3 without forming J densely at $O(QN^2L)$.

Proof. Write $R_i = \prod_{j\neq i} F_j$. For the own derivative, $\partial_{\mu_i} g_i(x) = -g_i'(x)$ (a location shift), and integrating by parts, $\int(-g_i')R_i\,dx = \int g_i R_i'\,dx = \int g_i R_i \sum_{j\neq i}\lambda_j\,dx = \int g_i R_i(\Lambda - \lambda_i)\,dx$. For $j \neq i$, $\partial_{\mu_j} F_j = -g_j$, so the cross derivative is $-\int g_i g_j \prod_{\ell\neq i,j} F_\ell\,dx = -\int g_i R_i \lambda_j\,dx$. Summing against h, $(Jh)_i = \int g_i R_i[\,h_i(\Lambda-\lambda_i) - \sum_{j\neq i} h_j\lambda_j\,]dx = \int g_i R_i(h_i\Lambda - A)\,dx$; average over f. $\square$

The differentiated map. Three maps must be distinguished: the exact max-wins map $p^{\max}(\mu)$; the reflected min-wins map $p^{\min}(a)$ used by the implementation, related by $a = -\mu$ so that $D_\mu p^{\max}[h] = D_a p^{\min}[-h]$; and the implemented approximation $p_{Q,L} = \tilde p_{Q,L}/(\mathbf{1}^\top \tilde p_{Q,L})$, whose derivative carries a normalization correction: with $v = D\tilde p[h]$ and $s = \mathbf{1}^\top\tilde p$,

$$Dp_{Q,L}[h] \;=\; \frac{v - p_{Q,L}\,(\mathbf{1}^\top v)}{s}.$$

Since $\mathbf{1}^\top v = 0$ analytically, the correction is of the order of the quadrature defect; empirically the uncorrected formula matches central finite differences of the returned normalized map to 5×10^{-9}. One further convention: the adaptive lattice endpoints depend on μ, so derivative statements refer to the *fixed-design* map: the lattice envelope and factor nodes are frozen during any inversion or outer-derivative computation. The inversion's analytic own-location slopes are accordingly a Jacobi quasi-Newton preconditioner, not the diagonal of the adaptive map's Jacobian.

Tail-stable implementation. Tail stability requires the log domain throughout: $\log\Phi$ evaluated directly (never $1-\Phi$, which underflows in double precision at $z \approx 8.3$), the analytic $\log g$, and hazards as $\exp(\log g - \log\Phi)$, which is bounded ($\sim z$ in the tail). Flooring a density before taking its logarithm while the exact log-survival stays finite produces spurious e^{+92} hazards on problems as mild as $N = 8$ with heterogeneous D; that configuration is a regression test. In the forward map the same discipline lets deep-tail shares resolve to genuine small positives rather than floored zeros.

Beyond Gaussian factors. The factors need not be Gaussian: (2) requires only conditional independence and an integrable factor law, so Student, mixture, or empirical factor distributions are admissible with appropriate nodes, and the base densities may be arbitrary and alternative-specific. One caution for asymmetric laws: the implementation works in the reflected (min-wins) race, and $-U$ has the same distribution as the reflected model only when the factor and base laws are symmetric; for asymmetric laws the reflected distributions must be used explicitly.

Numerical method. The benchmarks use $L = 1501$ lattice points on the adaptive interval $[\min_{i,q} m_i(f_q) - 8\max_i \sqrt{D_i},\ \max_{i,q} m_i(f_q) + 8\max_i \sqrt{D_i}]$, where $q = 1, \ldots, Q$ indexes the retained factor nodes.

Factor integration: the GHK benchmark (experiment 13) uses product Gauss–Hermite of order 15/11/9 for $k = 2/3/4$; the boundary experiment (experiment 14) uses order 15 per dimension for all $k \leq 4$, which after pruning weights below 10^{-7} of the maximum leaves $Q = 145$, 1317, and 10929 nodes for $k = 2, 3, 4$. The growth of Q with rank is exponential. Scrambled-Sobol nodes (2^{13} points, fixed seed) are used for $k > 4$: reproducible fixed-seed randomized QMC, disclosed as such.

Survival floors sit at 10^{-300} and the quadrature output is normalized by its sum. The returned map is a reproducible, piecewise-smooth numerical approximation of (2); statements about derivatives refer to derivatives of this approximation, whose fidelity to exact MNP derivatives is governed by L and Q.

Algorithm. Orientation: the implementation computes the reflected min-wins race on $a = -\mu$. Per retained factor node q: form conditional locations $m_i(f_q) = a_i + v_i^\top f_q$; on the uniform lattice evaluate $\log S_i = \log \Phi(-(x - m_i)/\sqrt{D_i})$; accumulate the field sum $\log S_{\text{field}} = \sum_j \log S_j$; recover each alternative's integrand as $g_i \exp(\log S_{\text{field}} - \log S_i)$; integrate by the rectangle rule with Δx weights and no endpoint modification (for integrands decaying to zero at both ends this equals the trapezoidal rule). Pruned factor weights are not renormalized explicitly; the deficit is absorbed by the final normalization $p = \tilde{p}/(\mathbf{1}^\top \tilde{p})$, and $|1 - \mathbf{1}^\top \tilde{p}|$ is reported as a diagnostic. Inversion: Newton step $\mu \leftarrow \mu - \text{damp} \cdot r/\partial_i$, with r the log-share residual on identified alternatives, ∂_i the analytic own-location log-slope, steps capped at $\sqrt{D_i + \|v_i\|^2}$, damping halved (floor 0.25) whenever the residual grows by more than 20%, tolerance 10^{-6} on the identified set, at most 50 iterations, followed by mean removal.

Inversion. Given target shares $\hat{p}$ and (V, D), the inverse transform finds mean-zero μ with $p(\mu) = \hat{p}$ by a damped Jacobi quasi-Newton iteration: all coordinates updated simultaneously with own-location slopes of the unnormalized lattice map as the preconditioner (available from the same pass), the field refrozen each iteration, a warm start from the independent-field inverse using total variances, and a tail-aware tolerance (alternatives with target shares below $10^{-4}/N$ are matched best-effort; their locations are weakly identified, which is also a conditioning caution for the reduced Jacobian when shares are tiny or alternatives nearly collinear). Convergence in a handful of iterations is an empirical observation on the tested problems, not a theorem. The design follows the original transform's inversion [10] and the calibration practice of [18].

2 Benchmark against GHK

Correctness anchors. On the binary case, where MNP has a closed form, the lattice transform agrees to 3.3×10^{-16}, and our GHK implementation (validated before being compared against) to 2×10^{-14} at $R = 2 \times 10^5$ (an easy anchor for both methods, useful for code validation rather than as evidence about the hard regime). At $N = 5$ both methods sit within the noise of a 10^7-draw Monte Carlo reference, and the shipped `thurstone.FactorRace` agrees with the benchmark kernel to 1.1×10^{-5}.

Error metric. Throughout, $\max_i |\hat{p}_i - p_i|$ against a simulation reference. A maximum over N coordinates tends to grow with N at fixed per-coordinate accuracy, and the reference's own maximum-coordinate noise behaves likewise; mean-absolute and total-variation summaries are committed alongside (experiment 16): at $N = 1000$ the lattice's mean absolute error is 5.6×10^{-6} and total variation 2.8×10^{-3} against a fresh 2×10^6-draw reference.

N	lattice transform	GHK ($R = 1000$)
5	20 ms, err 5.7×10^{-4}	1 ms, err 3.1×10^{-3}
20	86 ms, err 4.4×10^{-4}	12 ms, err 4.4×10^{-3}
50	217 ms, err 2.7×10^{-4}	77 ms, err 7.9×10^{-3}
200	860 ms, err 3.0×10^{-4}	1323 ms, err 6.8×10^{-3}
1000	4.3 s, err 6.8×10^{-4} †	—
5000	21.5 s, err 5.0×10^{-4} †	—

Table 1: Full share vector, $k = 2$ factors, one problem per size, generated as $\mu_i \sim N(0, s^2)$ with spread $s = 1$ through $N = 200$ and 1.5 above, $v_{ir} \sim N(0, 0.25/k)$, $D_i \sim U[0.5, 1.5]$. Sub-second timings are warmed medians of three runs. Reference: 2×10^6 MC draws through $N = 200$; 5×10^5 draws at $N = 1000$ and 5000 (†: there the reference's own maximum-coordinate noise is of comparable order, so these entries bound the lattice error only to that order). A replication study at common spread 1.0 (experiment 16; ten problems per size at $N \leq 200$, three at $N = 1000$, twin independent 2×10^6-draw references per problem) gives worst-case lattice error 3.5–4.6×10^{-4} at every size, at or below the references' own replicate-to-replicate noise. Lattice error stayed below 10^{-3} throughout; GHK cost crossed over by $N \approx 200$.

Matched-accuracy extrapolation. Fitting a power law to the reference implementation's timings over $N \in [20, 200]$ ($\alpha \approx 2.0$; a lower bound, the asymptotic per-share cost being cubic) and combining with $R^{-1/2}$ error scaling suggests that matching the lattice's 5×10^{-4} at $N = 5000$ would take this implementation on the order of fifty hours (0.27 h at $R = 1000$, times $(6.8 \times 10^{-3}/5 \times 10^{-4})^2 \approx 185$), against 21 seconds. The baseline is a straightforward single-threaded Python GHK: per-alternative Cholesky, plain pseudorandom uniforms, one realization per problem, shares normalized by their sum. It establishes the scaling of the per-alternative simulation approach, not a field-wide impossibility.

Direct simulation baseline. The strongest cheap baseline is *direct utility simulation*: draw utilities, take the argmax, average. It produces the whole share vector per draw and, unlike GHK, does not pay per alternative. We ran it at wall time matched to the lattice call (experiment 16): its maximum-coordinate error was 1.5–4× the lattice's at every tested N (6.2×10^{-4} versus 3.9×10^{-4} at $N = 1000$, 0.5M draws in the matched 4.1 seconds). At the 10^{-3} accuracy level, then, raw forward speed does not by itself separate the methods.

The separation is in error decay (experiment 17, $N = 200$, $k = 2$). Holding the factor rule fixed, lattice self-convergence is 2.7×10^{-8} at $L = 101$ and at machine precision from $L \approx 200$: the integrand is analytic and endpoint-decaying, so the uniform lattice is effectively spectrally accurate. L-error is subdominant. Holding L fixed, the Gauss–Hermite sweep decays from 5.3×10^{-4} (order 3) to 5.9×10^{-9} (order 15). The pre-normalization defect $|1 - \mathbf{1}^\top \tilde{p}\,|$ is 2.4×10^{-8}.

On the same problem the measured accuracy–time frontier (Figure 2, left) has the lattice at 4.1×10^{-5} error in 0.03 seconds (at the twin-reference noise floor), direct MC at 1.2×10^{-4} in 15.7 seconds ($R = 10^7$), scrambled-Sobol direct QMC at 2.0×10^{-4} in 2.6 seconds (2^{20} points), GHK at 2.0×10^{-3} in 10.7 seconds ($R = 10^4$; over 8 seeds at $R = 10^3$ its error spans $[3.9\times10^{-3}, 1.3\times10^{-2}]$), and QMC-GHK (scrambled-Sobol uniforms in the same kernel) at 4.2×10^{-4} in 7.0 seconds ($R = 8192$), a five-to-sevenfold error improvement over plain GHK at equal R and still well off the lattice frontier. Below 10^{-3} the deterministic map separates decisively, consistent with $R^{-1/2}$ against a rapidly converging quadrature. One stronger stochastic baseline remains to be added: minimax tilting [9].

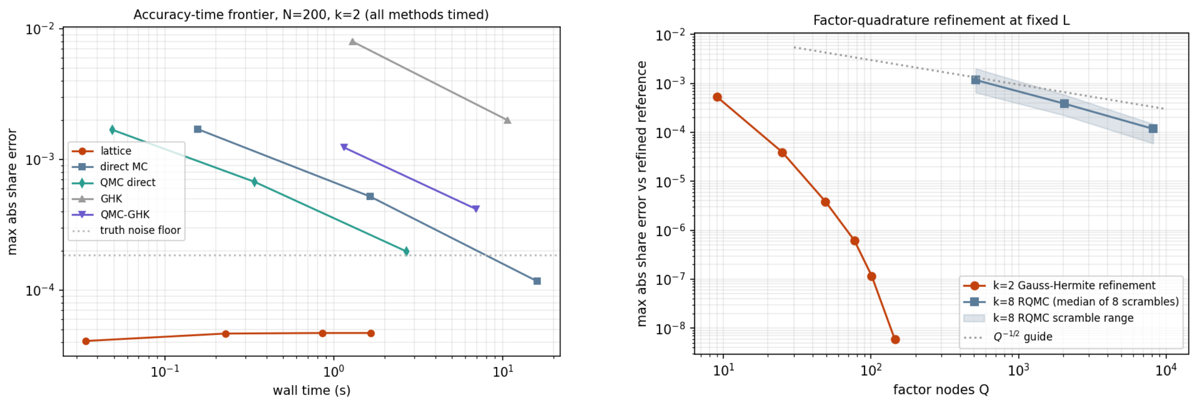

Figure 2: Left: measured accuracy–time frontier at $N = 200$, $k = 2$ (experiment 17); every method is timed on the same problem against twin 5×10^7-draw references, with direct simulation and QMC-GHK as first-class baselines. Right: factor-quadrature refinement at fixed L: Gauss–Hermite at $k = 2$, and at $k = 8$ the median and range over eight independent RQMC scrambles.

Derivatives. For fixed factor nodes and a fixed lattice, the transform is a reproducible, (piecewise-)differentiable function of μ whose derivatives are evaluated without resampling; analytic own-location slopes of the unnormalized map come from the same pass and serve as the inversion preconditioner. (Along a utility path at $N = 50$, the maximum second difference over dt^2 is 39.3 for fresh-draw GHK against 0.01 for both CRN-GHK and the lattice; the committed smoothness figure shows the curves.) These are derivatives of the numerical approximation, not exact MNP derivatives, and GHK-based practice likewise differentiates its approximation, analytically or with fixed draws [3, 4]. The practical distinction is the character and controllability of the approximation error, quadrature resolution versus draw count, not the possibility of differentiation.

3 Calibration at $N = 1000$

At $N = 1000$, $k = 2$, with targets from 5×10^6 MC draws floored at 10^{-7} and renormalized: the inversion completes in 57 seconds. The iteration solves its own equations to 6.4×10^{-9} (a statement about the solver, not statistical accuracy, since the target carries sampling noise near 2×10^{-4}) and recovers the utilities of identified alternatives (target share above 3×10^{-4}) to maximum error 0.017. A replication study (experiment 16) repeats the inversion on a second

$N = 1000$ problem against three independent 5×10^6-draw targets: 57 seconds each, 161–162 alternatives identified (98.1% of share mass), recovery error maximum 0.015–0.023 and median ≈ 0.0027 across replicates, with the recovery-versus-share curve committed alongside. Numerical well-posedness matches Theorem 1: the reduced Jacobian assembled from N JVP calls at $N = 50$ is symmetric to 1.7×10^{-18}, has row-sum defect 6.2×10^{-17}, and reduced eigenvalues in $[2.5 \times 10^{-5}, 0.22]$, positive definite as proved. We are not aware of probit share inversion at this scale in current simulation-based practice.

Removal ensemble (by-product). The cached field also yields all N single-removal share vectors in $O(QN^2L)$. Against the strongest simulation comparator, which reuses each draw's winner and runner-up to answer every removal at once in $O(RN+N^2)$, the deterministic ensemble is about 3× more accurate at matched wall time at $N = 200$ (4.2×10^{-5} versus 1.2×10^{-4}; experiment 18). This is a side capability, not the paper's case.

4 Boundary experiments

General covariance through the factor approximation. The method requires $\Sigma \approx VV^\top + D$; GHK does not. Choices depend on (μ, Σ) only through $P\mu$ and $P\Sigma P$, $P = I - \mathbf{1}\mathbf{1}^\top/N$: a common factor loading $V \to V + \mathbf{1}c^\top$ shifts every conditional location equally and cannot move the argmax (a machine-precision identity), so the factor fit must be run in this choice-relevant quotient space. Fitting raw Σ can spend scarce rank on an irrelevant common component ($\Sigma = \tau^2\mathbf{1}\mathbf{1}^\top + \beta\beta^\top + D$ with large τ: the raw rank-1 fit takes $\mathbf{1}\mathbf{1}^\top$ and misses $\beta\beta^\top$ entirely; on such an example the contrast-space fit cuts rank-1 share error from 4.6×10^{-2} to 8.3×10^{-3}). The asymmetry matters: only the common *factor* direction is irrelevant; a common addition to D inflates every difference variance and is choice-relevant, so D must come from the quotient fit. Loadings are centered and SVD-canonicalized (ordered columns, fixed signs), making results reproducible at the covariance level.

With this fit: correlation matrices at spectral decay $\lambda_m \propto m^{-\gamma}$, $\gamma \in \{0.5, 1.5, 3\}$, at $N = 50$, sharing one orthogonal eigenbasis across γ so decay comparisons are not confounded with basis realization. The power law holds before diagonal standardization, which perturbs the spectrum; the actual leading eigenvalues (3.7, 17.1, 34.2 at $\gamma = 0.5, 1.5, 3$) are printed by the script. Share error is measured against an 8×10^6-draw reference, with GHK at the exact Σ for comparison; the plotted quantity is the rank-k approximation error of the stated procedure, not a minimum over factor models. Rank-8 errors: 1.8×10^{-3}, 3.7×10^{-3}, and 3.1×10^{-3} at $\gamma = 0.5, 1.5, 3$ from $k = 1$ starting points of 2.3×10^{-3}, 1.5×10^{-2}, 7.1×10^{-2}.

Decomposing the error with a second simulation under the fitted factor model: integration error is flat at 2–7×10^{-4} across every (γ, k) tested, so the rank-k error is essentially covariance-fit error. Rank, not quadrature, is the binding constraint. Against GHK at $R = 10^4$ (3.2, 2.3, 4.3×10^{-3}), rank-8 wins at $\gamma = 0.5$ and 3 but *loses at intermediate decay* $\gamma = 1.5$ (3.7×10^{-3} versus 2.3×10^{-3}): a GHK-wins regime does exist, where the spectrum is too flat to truncate and too structured to ignore. Nor is this wall-time-matched: the $k = 8$ RQMC evaluation takes 15–24 s against 0.5–0.9 s for GHK at $R = 10^4$, so at $N = 50$ GHK is the cheaper route to 3–4×10^{-3}; the lattice's advantage is large N at small k. Where accuracy demands exceed the achievable rank-k error, GHK's arbitrary-Σ generality is the right tool.

Substitution: a 2×2 factorial. The design varies factor structure and noise family one axis at a time, holding the idiosyncratic scale common. Coupling either axis with alternative-specific scales would confound the comparison: an alternative-specific-scale Gumbel race is not mixed logit, and heteroskedasticity alone already breaks IIA. A separate heteroskedasticity axis is left to future replication.

Ground truth is misspecified for every candidate: $t(5)$ factors and skew-normal idiosyncratic noise standardized to mean zero and unit variance, homoskedastic. (The race is computed min-wins, so positive skew in race coordinates is *left*-skew in utility coordinates.) The four candidates hold the idiosyncratic scale fixed at one and vary $\{\text{Gumbel}, \text{Gaussian}\}$ base $\times$ $\{V = 0, V = V^\star\}$: independent Luce (= plain multinomial logit exactly), independent probit, factor mixed logit (common-scale), and factor probit. All are calibrated to the same full-menu shares (residuals $\leq 4 \times 10^{-10}$).

Deletion truths come from 2×10^7 *common* utility draws with the top three alternatives retained per draw, from which every single-deletion and pair-deletion truth follows with common random numbers: 105 informative blocks stratified by deleted mass (26 / 29 / 28 blocks at >10% / 2–10% / 0.5–2%). The score per deletion block B is the misallocated fraction of the redistributed mass,

$$\mathrm{TV}\Big(q^{(-B)}_{\text{model}},\, q^{(-B)}_{\text{truth}}\Big) \Big/ \textstyle\sum_{i \in B} p_i,$$

averaged within strata:

	mass > 10%	2–10%	0.5–2%
independent Luce (= plain logit)	0.229	0.245	0.261
independent probit	0.210	0.232	0.230
factor mixed logit	0.114	0.146	0.199
factor probit	**0.045**	**0.062**	**0.094**

The factorial decomposition (top two strata): the factor increment is dominant in both families (+0.115/+0.099 within Gumbel, +0.165/+0.170 within Gaussian); the family increment alone is small ($+0.019/+0.013$ at $V = 0$) but grows with factors present ($+0.069/+0.084$), a negative interaction ($-0.050/-0.071$): factor structure and the Gaussian base are complements in this design. Caveats: a single truth design and seed, and a skew-normal truth that is Gaussian-adjacent. The figure shows every block individually.

5 Related work

GHK and simulated likelihood are standard [2, 3, 4]. Bhat's MACML [7] and the approximations surveyed by Connors, Hess and Daly [8] trade simulation for analytic marginals, still one alternative at a time. Botev [9] sharpened Gaussian-restriction simulation with minimax tilting.

Factor dimension reduction for probit is old. Butler and Moffitt [5] integrated one factor by quadrature; Elrod and Keane [6] estimated factor-analytic probit for panel data; Loaiza-Maya and Nibbering [17] pursue large choice sets by Bayesian estimation. The lattice survival-product identity and its inverse transform come from racing [10].

Inversion theory is likewise established. Norets and Takahashi [16] prove surjectivity of the utility-to-probability map; Chiong, Galichon and Shum [15] invert through the convex conjugate; Berry, Levinsohn and Pakes [11] built demand estimation on the logit-side inversion, and

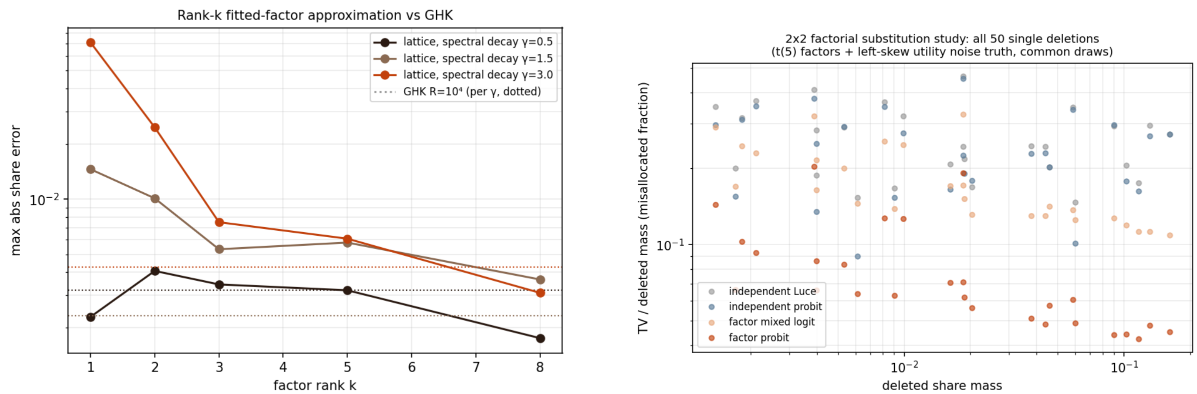

Figure 3: Left: rank-k contrast-space approximation error under spectral decay γ (dotted: GHK at $R = 10^4$, per γ; not wall-time matched). Right: the 2×2 factorial substitution study, with every informative single-deletion block shown individually.

Conlon and Gortmaker [12] maintain its modern practice. Chia [13] differentiates logit BLP automatically; Huch and Keane [14] amortize correlated-choice probabilities with neural networks, evidence of current demand for this computation.

The novelty here is accordingly none of: dimension reduction, the order-statistic integral, or well-posedness of inversion. It is the all-alternatives assembly in $O(QNL)$, its large-N implementation, the $O(QNL)$ Jacobian–vector product, and the shared-field reuse for calibration and counterfactuals.

6 Discussion

Estimation is the natural next step. With the reduced coordinates of Proposition 3, gradients of an outer likelihood or GMM objective pass through the calibration fixed point by implicit differentiation, a route to differentiable probit demand estimation at large N. Estimating (V, D) rather than supplying them raises the usual identification cautions: scale normalization, factor rotation, and the fact that only difference covariances are identified.

Beyond the Gaussian family, the same machinery computes races with arbitrary alternative-specific idiosyncratic densities. A common-scale Gumbel base with zero loadings reproduces the Luce model exactly, yielding correlated-softmax generalizations.

Neither method class dominates the other. Simulation owns arbitrary Σ and coarse accuracy; the shared field owns the factor form, tight accuracy, derivatives, and calibration; the boundary experiments draw the border at intermediate spectral decay. We leave to future work the minimax-tilting baseline, replication of the substitution factorial across truth designs, and a walk-forward calibration on real assortment data.

References

[1] L. L. Thurstone. A law of comparative judgment. *Psychological Review* 34 (1927) 273–286.

[2] J. Geweke, M. Keane, D. Runkle. Alternative computational approaches to inference in the multinomial probit model. *Review of Economics and Statistics* 76 (1994) 609–632.

[3] V. Hajivassiliou, D. McFadden. The method of simulated scores for the estimation of LDV models. *Econometrica* 66 (1998) 863–896.

[4] K. Train. *Discrete Choice Methods with Simulation*, 2nd ed. Cambridge, 2009.

[5] J. S. Butler, R. Moffitt. A computationally efficient quadrature procedure for the one-factor multinomial probit model. *Econometrica* 50 (1982) 761–764.

[6] T. Elrod, M. P. Keane. A factor-analytic probit model for representing the market structure in panel data. *Journal of Marketing Research* 32 (1995) 1–16.

[7] C. Bhat. The maximum approximate composite marginal likelihood (MACML) estimation of multinomial probit-based unordered response choice models. *Transportation Research B* 45 (2011) 923–939.

[8] R. Connors, S. Hess, A. Daly. Analytic approximations for computing probit choice probabilities. *Transportmetrica A* 10 (2014) 119–139.

[9] Z. Botev. The normal law under linear restrictions: simulation and estimation via minimax tilting. *JRSS-B* 79 (2017) 125–148.

[10] P. Cotton. Inferring relative ability from winning probability in multientrant contests. *SIAM J. Financial Mathematics* 12 (2021) 295–317.

[11] S. Berry, J. Levinsohn, A. Pakes. Automobile prices in market equilibrium. *Econometrica* 63 (1995) 841–890.

[12] C. Conlon, J. Gortmaker. Best practices for differentiated products demand estimation with pyblp. *RAND Journal of Economics* 51 (2020) 1108–1161.

[13] A. Chia. Automatically differentiable random coefficient logistic demand estimation. arXiv:2106.04636 (2021).

[14] E. Huch, M. Keane. Amortized inference for correlated discrete choice models via equivariant neural networks. arXiv:2603.24705 (2026).

[15] K. Chiong, A. Galichon, M. Shum. Duality in dynamic discrete-choice models. *Quantitative Economics* 7 (2016) 83–115.

[16] A. Norets, S. Takahashi. On the surjectivity of the mapping between utilities and choice probabilities. *Quantitative Economics* 4 (2013) 149–155.

[17] R. Loaiza-Maya, D. Nibbering. Scalable Bayesian estimation in the multinomial probit model. *Journal of Business & Economic Statistics* 40 (2022) 1678–1690.

[18] The `allocation` package: Thurstone-tilt portfolio construction. `https://github.com/microprediction/allocation`.